

UNIT–1: Introduction to Artificial Intelligence

Artificial Intelligence (AI)

Artificial Intelligence (AI) is a branch of computer science that focuses on creating machines and computer systems that can **perform tasks which normally require human intelligence**.

These tasks include **learning, reasoning, problem-solving, understanding language, recognizing images, making decisions, and understanding the environment**.

Simple Definition:

AI is the ability of a computer or machine to think, learn, and make decisions in a way that resembles human intelligence.

Examples of AI

- **ChatGPT** – understands and generates human language.
- **Google Maps** – finds routes and predicts traffic.
- **Face Recognition** – identifies people from images.
- **Voice Assistants** – understand and respond to spoken commands.
- **Recommendation Systems** – suggest videos, products, or songs.
- **Self-driving Cars** – understand roads and make driving decisions.

What is Human Intelligence?

Before understanding Artificial Intelligence, students should understand **intelligence**.

Human intelligence allows us to:

- Learn from experience
- Understand language
- Recognize objects
- Solve problems
- Make decisions
- Understand patterns
- Adapt to new situations
- Reason and analyze information

For example:

When a person sees a dog, the human brain can identify it as a dog based on previous knowledge and experience.

AI tries to enable computers to perform similar types of tasks.

Human Intelligence vs Artificial Intelligence

Human Intelligence	Artificial Intelligence
Natural and biological	Artificial and computer-based
Uses the human brain	Uses algorithms and computational models

Learns from experience and environment	Learns mainly from data/training
Has emotions and empathy	Does not naturally have human emotions
Strong common-sense understanding	Limited common-sense understanding
Generally flexible across many situations	Often specialized for specific tasks
Can learn from very few examples	Often requires substantial data
Creativity is naturally present	Can generate creative-looking outputs but works through computational processes

Conclusion

Human intelligence is **biological and naturally developed**, while AI is **artificially designed to perform intelligent tasks**.

Why Do We Need Artificial Intelligence?

AI is useful because computers can process huge amounts of data much faster than humans.

AI can be used to:

- Automate repetitive tasks
- Analyze large amounts of data
- Make predictions
- Detect patterns
- Provide recommendations
- Understand human language
- Recognize images and speech
- Assist humans in decision-making

Example

Consider an online shopping website.

It may analyze:

- What products you searched for
- What products you purchased
- What products you viewed
- What products similar users purchased

Based on these patterns, AI can recommend products.

Scope of Artificial Intelligence

The **scope of AI** refers to the areas where AI can be applied to solve problems, automate tasks, and improve decision-making.

Area	Application of AI
Healthcare	Disease detection, medical diagnosis, drug discovery
Education	AI tutors, personalized learning, automatic evaluation
Banking	Fraud detection, credit scoring, chatbots

Business	Customer support, data analysis, prediction
Transportation	Autonomous vehicles, route optimization
Agriculture	Crop monitoring, disease detection, smart irrigation
Cybersecurity	Threat detection and anomaly detection
Entertainment	Movie, music and video recommendations
Robotics	Industrial robots and service robots
Natural Language	Translation, chatbots, speech recognition
Computer Vision	Face recognition, object detection, image analysis

Future Scope of AI

AI is expected to become increasingly important in:

Healthcare → Education → Robotics → Autonomous Systems → Cybersecurity → Scientific Research → Smart Cities → Business Automation

However, AI also creates challenges such as **privacy, bias, security, job displacement, misinformation, and ethical use.**

Brief History of Artificial Intelligence

The development of AI has taken place over several decades. Some important milestones are:

1. Alan Turing – 1950

British mathematician **Alan Turing** was one of the earliest thinkers in the field of machine intelligence.

In 1950, he published the paper “**Computing Machinery and Intelligence.**” He proposed the famous **Turing Test**, which asks whether a machine can communicate in such a way that a human cannot reliably distinguish it from another human.

Importance: Turing provided an early theoretical foundation for thinking about machine intelligence.

2. Birth of AI – 1956

The term “**Artificial Intelligence**” was introduced by **John McCarthy** in 1956 at the **Dartmouth Conference**.

Researchers proposed that machines could be made capable of performing tasks associated with human intelligence.

John McCarthy → Father of Artificial Intelligence

3. Early AI Programs – 1950s–1960s

Researchers developed programs capable of solving mathematical problems, playing games, and performing logical reasoning.

Examples included:

- **Logic Theorist (1956)** – developed by Allen Newell, Herbert Simon and Cliff Shaw.
- **ELIZA (1966)** – an early chatbot developed by Joseph Weizenbaum.

These systems were mainly **rule-based** and worked within limited environments.

4. Expert Systems – 1970s–1980s

AI research moved toward **Expert Systems**.

An expert system uses a collection of rules and knowledge to make decisions in a specific domain.

Example:

A medical expert system could use symptoms and medical rules to suggest possible diseases.

AI became useful in areas such as **medicine, finance, and industrial applications**.

5. AI Winter – 1970s and 1980s

AI research faced periods known as **AI Winters**, when funding and interest decreased.

The major reasons included:

- Limited computing power
- Lack of sufficient data
- High expectations from AI
- Difficulty in solving complex real-world problems

6. Machine Learning Growth – 1990s–2000s

AI gradually shifted from manually programmed rules toward **Machine Learning (ML)**.

Instead of explicitly programming every rule, machines could **learn patterns from data**.

A famous milestone was **IBM Deep Blue defeating chess champion Garry Kasparov in 1997**.

7. Deep Learning Revolution – 2010s

With the availability of:

Big Data + Powerful GPUs + Better Algorithms

deep learning became highly successful.

Deep Neural Networks achieved major improvements in:

- Image recognition
- Speech recognition
- Natural Language Processing
- Computer Vision

AlexNet (2012) was a major milestone in the development of modern deep learning.

8. Generative AI – 2020s

AI systems started becoming capable of **generating new content**, rather than only classifying or predicting existing information.

Generative AI can create:

- Text
- Images
- Audio
- Video
- Computer Code

Examples include **ChatGPT, Gemini, Claude, and image-generation models.**

9. ChatGPT – 2022

ChatGPT was publicly launched by OpenAI in **November 2022.**

It is based on **Generative AI and Large Language Model (LLM)** technology.

ChatGPT can:

- Understand natural-language questions
- Generate human-like text
- Explain concepts
- Summarize information
- Write and debug code
- Assist in problem-solving
- Generate ideas and content

ChatGPT significantly increased public awareness and practical use of **Generative AI.**

Types of Artificial Intelligence

AI can be classified mainly on the basis of its **capabilities and level of intelligence** into three types:

1. **Narrow AI (Weak AI)**
2. **General AI (Strong AI)**
3. **Super AI (Artificial Superintelligence)**

1. Narrow AI (Weak AI)

Narrow AI is an AI system designed to perform a **specific task or a limited set of related tasks.**

It can perform its assigned task very efficiently, but it **cannot independently perform every type of intellectual task that a human can do.**

Examples:

- ChatGPT
- Google Maps
- Face recognition
- Voice assistants such as Siri
- Recommendation systems of YouTube/Netflix
- Spam detection

Key Point:

Narrow AI is specialized in a particular task and cannot automatically transfer its intelligence to completely different tasks.

Current Status: Widely available and commonly used.

2. General AI (AGI)

Artificial General Intelligence (AGI) refers to a hypothetical AI system that would have **human-like general intelligence**.

An AGI system would be able to **learn, reason, understand, solve problems, and apply knowledge across many different domains**, rather than being limited to one specific task.

Example:

Imagine a machine that can learn mathematics, teach a class, write software, understand a new language, drive a car, and solve an unfamiliar problem without being separately designed for each task.

Key Point:

AGI aims to perform a wide range of intellectual tasks at a level comparable to humans.

Current Status: Not yet achieved.

3. Super AI

Super AI, also called **Artificial Superintelligence (ASI)**, refers to a hypothetical AI whose intelligence would **surpass human intelligence across virtually all intellectual areas**.

It could potentially perform tasks such as reasoning, scientific research, creativity, strategic planning, and problem-solving at a level far beyond humans.

Example:

A hypothetical AI capable of discovering new scientific theories, solving extremely complex global problems, and outperforming the best human experts in almost every intellectual field.

Current Status: Hypothetical / does not currently exist.

Types of AI — Quick Comparison

Type	Intelligence	Ability	Status
Narrow AI	Task-specific	Performs specific tasks	Exists
General AI (AGI)	Human-level general intelligence	Performs many different intellectual tasks	Not achieved
Super AI (ASI)	Beyond human intelligence	Outperforms humans in almost all intellectual tasks	Hypothetical

AI vs Machine Learning vs Deep Learning

These three terms are related, but they are **not the same**.

The easiest way to understand their relationship is:

Artificial Intelligence (AI)



Machine Learning (ML)



Deep Learning (DL)

In simple words:

AI is the broad concept → ML is a way to achieve AI → DL is a specialized approach within ML.

Artificial Intelligence (AI)

Artificial Intelligence is the broad field of computer science concerned with creating systems that can perform tasks requiring human-like intelligence.

AI may use different techniques such as:

- Rule-based systems
- Search algorithms
- Machine Learning
- Deep Learning
- Natural Language Processing
- Computer Vision
- Robotics

Example: A chess-playing system that can make intelligent decisions.

Machine Learning (ML)

Machine Learning is a subset of AI in which computers **learn patterns from data and improve their performance without being explicitly programmed for every rule.**

Instead of giving the computer all the rules manually, we provide **data and an appropriate learning algorithm**, and the system learns patterns from that data.

Example:

Give an ML system thousands of emails labeled **Spam** and **Not Spam**. It learns patterns and can then predict whether a new email is spam.

Main Types of ML

- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning

Deep Learning (DL)

Deep Learning is a subset of Machine Learning that uses **artificial neural networks with multiple layers**, called **deep neural networks**, to learn complex patterns from large amounts of data.

Deep Learning is particularly successful in:

- Image recognition
- Speech recognition
- Natural Language Processing

- Computer Vision
- Generative AI

Example:

A deep learning model can analyze millions of images and learn to recognize objects such as **cats, dogs, cars, or faces**.

Difference Between AI, ML and Deep Learning

Basis	AI	Machine Learning	Deep Learning
Meaning	Broad field of intelligent machines	Method of learning from data	Advanced ML using deep neural networks
Scope	Broadest	Subset of AI	Subset of ML
Main Idea	Make machines intelligent	Machines learn from data	Machines learn complex patterns using neural networks
Data Requirement	May or may not require large data	Generally requires data	Usually requires large amounts of data
Human Intervention	Can use manually defined rules	Less manual rule creation	Learns features automatically
Algorithms	Rules, search, ML, DL, etc.	Decision trees, regression, SVM, etc.	CNN, RNN, Transformers, etc.
Computational Power	Depends on technique	Moderate to high	Usually high
Example	Intelligent robot	Spam prediction	Face/image recognition

AI Hallucination

One of the most important limitations of Generative AI is **hallucination**.

AI may sometimes generate information that:

- Sounds correct
- Looks confident
- But is actually incorrect or fabricated

Example

A student asks:

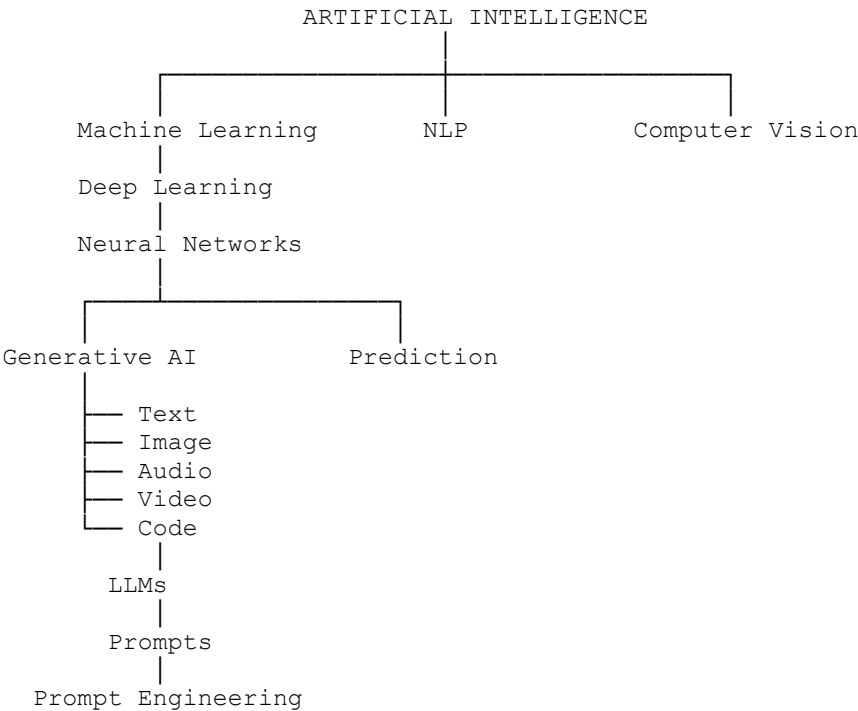
Who invented a fictional technology?

The AI may sometimes generate a convincing but incorrect answer.

Therefore:

AI-generated information should be verified, especially for academic, technical, legal, medical, financial, and other high-stakes information.

Complete AI Concept Map



How Does a Machine Learn from Data?

Machine Learning does **not mean that a machine thinks exactly like a human**. It means that a computer algorithm analyzes data, finds useful **patterns**, and uses those patterns to make **predictions or decisions** on new data.

The basic process is:

Data → Pattern Identification → Model Training → Prediction

1. Data

Data is the information from which a machine learns.

Data can be:

- Numbers
- Text
- Images
- Audio
- Videos
- Sensor readings

Example: Student Performance

Suppose we want to predict whether a student will **pass or fail** based on study hours.

Study Hours	Attendance	Result
2	60%	Fail
3	65%	Fail
4	70%	Pass

5	75%	Pass
6	80%	Pass
7	85%	Pass

Here:

- **Study Hours and Attendance → Features/Input**
- **Result → Target/Output**

The machine is given these examples during **training**.

2. Finding Patterns

After receiving the data, the ML algorithm tries to find a **relationship or pattern** between the input and output.

From the above data, the machine may discover a pattern such as:

Generally, students who study more hours and have better attendance are more likely to pass.

The machine does not simply memorize one student's result. It tries to learn a **general relationship** from many examples.

For example:

More Study Hours + Better Attendance → Higher Probability of Passing

This learned relationship is called a **model**.

3. Training the Model

The process of allowing an ML algorithm to learn patterns from data is called **training**.

During training:

Training Data → ML Algorithm → Learning Patterns → Trained Model

For example, the algorithm may initially make incorrect predictions:

4 hours → Fail

It compares its prediction with the actual result:

Actual result → Pass

The algorithm adjusts its internal parameters to reduce the error.

This process is repeated many times until the model learns a useful relationship.

Simple Example

Suppose the model learns approximately:

If study hours and attendance are sufficiently high, the probability of passing increases.

The actual mathematical relationship depends on the algorithm being used.

4. Making Predictions

Once the model has learned from the training data, we can provide **new/unseen data**.

Suppose a new student has:

Study Hours = 5

Attendance = 78%

The model uses the patterns it learned during training and predicts:

Prediction: Pass

The important point is that this student's data was **not necessarily present in the training examples**.

This is the main purpose of Machine Learning: **learning from previous examples and applying that learning to new data**.

5. Real-Life Example: Spam Email Detection

Consider an email system that wants to identify whether an email is **Spam or Not Spam**.

Step 1: Collect Data

The system receives thousands of previously classified emails.

Email Content	Label
"Congratulations! You won a prize."	Spam
"Click here to claim your reward."	Spam
"Meeting scheduled at 10 AM."	Not Spam
"Please send the project report."	Not Spam

Here, the email text is the **input data**, and Spam/Not Spam is the **target**.

Step 2: Find Patterns

The ML algorithm may discover that spam emails frequently contain patterns such as:

- "Congratulations"
- "Win"
- "Prize"
- "Free"
- Suspicious links
- Repeated promotional language

It doesn't mean that every email containing "free" is spam. The model learns from **many features together**.

Step 3: Train the Model

The algorithm adjusts its internal parameters based on thousands or millions of examples.

Emails → ML Algorithm → Learned Patterns → Spam Detection Model

Step 4: Predict a New Email

Now a new email arrives:

"Congratulations! You have won a free prize. Click this link."

The trained model analyzes the email and predicts:

Spam → 98% probability

The exact probability depends on the trained model and data.

6. Another Example: House Price Prediction

Suppose we want to predict the price of a house.

We collect historical data:

Area (sq ft)	Bedrooms	Location	Price
800	2	City A	₹40 Lakh
1000	2	City A	₹50 Lakh
1200	3	City A	₹60 Lakh
1500	3	City A	₹75 Lakh
1800	4	City A	₹90 Lakh

The machine may learn patterns such as:

Larger Area + More Bedrooms + Better Location → Generally Higher Price

Now suppose a new house has:

Area = 1300 sq ft

Bedrooms = 3

Location = City A

The trained model might predict:

Estimated Price ≈ ₹65 Lakh

This is a **prediction**, not a guarantee.

How Does the Machine Actually "Learn"?

A computer does not learn like a human brain by simply understanding the meaning of the data. Instead, an **ML algorithm mathematically adjusts parameters** inside a model so that its predictions become closer to the correct answers.

The simplified process is:

Input Data



ML Algorithm



Initial Prediction



Compare with Actual Answer



Calculate Error



Adjust Model Parameters



Repeat



Trained Model

This process is called **model training**.

Important Terms

Data: Information used by an ML system for learning.

Feature: An input characteristic used for making a prediction, e.g., study hours, age, area of a house.

Target/Label: The desired output, e.g., Pass/Fail or House Price.

Pattern: A relationship or regularity discovered in the data.

Training: The process of learning patterns from training data.

Model: The learned mathematical representation used to make predictions.

Prediction: The output generated by the trained model for new/unseen data.

AI in Daily Life

Artificial Intelligence is no longer limited to research laboratories. **AI is already a part of many applications that we use in our daily lives.** It works behind the scenes to analyze data, identify patterns, make predictions, and provide personalized results.

1. Google Maps

AI is used in **Google Maps** to analyze location, traffic, and route-related data.

AI helps in:

- Predicting traffic conditions
- Finding the fastest route
- Estimating **time of arrival (ETA)**
- Suggesting alternative routes
- Predicting busy areas

Example: If Google Maps detects heavy traffic on your usual route, it can predict a delay and suggest another route.

AI Concept: Prediction + Pattern Recognition

2. Spotify

Spotify uses **AI and Machine Learning** to provide personalized music recommendations.

It can analyze:

- Songs you listen to
- Songs you skip
- Artists you follow
- Listening history
- Preferences of similar users

Example: If you frequently listen to a particular singer or genre, Spotify may recommend similar songs and artists.

AI Concept: Recommendation System

3. Netflix

Netflix uses **AI and Machine Learning** to recommend movies and TV shows according to a user's interests.

It can consider:

- Watch history
- Preferred genres
- Movies/shows you complete or stop watching
- Ratings and interactions
- Viewing patterns of similar users

Example: If you frequently watch action and thriller movies, Netflix may recommend more content from these categories.

AI Concept: Personalization + Recommendation

4. Google Photos

Google Photos uses **AI and Computer Vision** to understand the content of images.

AI can help identify:

- People and faces
- Objects
- Places
- Animals
- Text in images
- Different scenes

Example: If you search for "**dogs**" in Google Photos, it can find photos containing dogs without manually adding a "dog" label to every photo.

AI Concept: Computer Vision + Image Recognition

5. Voice Assistants

Voice assistants such as **Google Assistant, Siri, and Alexa** use AI to understand spoken language and respond to users.

The basic process is:

Speech → Speech Recognition → Natural Language Processing → Response

Example:

You say: "Set an alarm for 7 AM."

The system recognizes your speech, understands your command, and creates the alarm.

AI Concepts: Speech Recognition + Natural Language Processing (NLP)

Quick Table

Application	How AI is Used	Main Technology
Google Maps	Traffic and route prediction	Prediction, ML
Spotify	Music recommendations	Recommendation System
Netflix	Movie/show recommendations	Recommendation System
Google Photos	Recognizes faces, objects, and scenes	Computer Vision
Voice Assistants	Understands and responds to speech	NLP + Speech Recognition

Machine Learning Concepts and Learning Strategies

1. What is Machine Learning?

Machine Learning (ML) is a subset of Artificial Intelligence in which a computer **learns patterns from data and uses those patterns to make predictions or decisions** without being explicitly programmed for every situation.

Basic Process

Data → Learning Algorithm → Pattern → Model → Prediction/Decision

Example

Suppose we provide a machine with thousands of emails labeled **Spam** and **Not Spam**. The machine analyzes these examples and learns patterns commonly found in spam emails. When a new email arrives, the trained model predicts whether it is **Spam or Not Spam**.

Learning Strategies in Machine Learning

Machine Learning is mainly divided into three learning strategies:

1. **Supervised Learning**
2. **Unsupervised Learning**
3. **Reinforcement Learning**

The major difference is **how the machine receives feedback or information during learning**.

1. Supervised Learning

Supervised Learning is a type of Machine Learning in which the model learns from **labeled data**.

Here, both the **input and the correct output are provided** during training. The model learns the relationship between them and uses it to predict the output for new data.

Process

Input + Correct Output → Training → Learn Relationship → New Input → Prediction

Example: Student Result Prediction

Study Hours	Result
2	Fail
3	Fail
5	Pass
6	Pass
8	Pass

The machine learns the relationship between **study hours** and **result**.

Now, if a new student studies for **7 hours**, the model may predict:

Prediction → Pass

Applications

- Spam email detection
- House price prediction
- Disease prediction
- Student result prediction
- Face recognition

Main Types

A. Classification: Output belongs to a category.

Example:

Email → Spam / Not Spam

B. Regression: Output is a numerical value.

Example:

House features → ₹50 lakh

Key Point

Supervised Learning = Learning with labelled data and known answers.

2. Unsupervised Learning

Unsupervised Learning is a type of Machine Learning in which the model learns from **unlabelled data**.

Here, the correct output is **not provided**. The machine tries to discover **hidden patterns, structures, or groups** in the data on its own.

Process

Unlabelled Data → Algorithm → Find Patterns/Groups → Useful Information

Example: Customer Segmentation

Suppose a shopping company has information about thousands of customers:

- Age
- Spending
- Number of purchases
- Product preferences

But there is **no predefined category** such as "Premium Customer" or "Budget Customer."

An unsupervised learning algorithm can identify groups such as:

Group 1 → High spending customers

Group 2 → Medium spending customers

Group 3 → Low spending customers

The machine discovered these groups from the data itself.

Applications

- Customer segmentation
- Grouping similar documents
- Market analysis
- Anomaly detection
- Finding hidden patterns in data

Common Technique

Clustering is one of the most common techniques in unsupervised learning.

Examples:

- K-Means Clustering
- Hierarchical Clustering

Key Point

Unsupervised Learning = Learning from data without predefined labels or answers.

3. Reinforcement Learning

Reinforcement Learning (RL) is a learning strategy in which an **agent learns by interacting with an environment** and receiving **rewards or penalties** for its actions.

The goal of the agent is to learn a strategy that **maximizes the total reward over time**.

Important Components

Agent: The learner or decision-maker

Environment: The world in which the agent operates

Action: What the agent does
Reward: Feedback received after an action
State: Current situation of the environment

Basic Process

State → Agent → Action → Environment → Reward → New State → Repeat

Example: Game Playing

Consider an AI learning to play a game.

- AI moves the game character → **Action**
- Character reaches the goal → **+10 Reward**
- Character loses → **-10 Penalty**
- AI repeats the game many times.
- Gradually, it learns which actions produce better results.

The AI does not receive a complete list of correct moves. It **learns through trial and error**.

Real-Life Applications

- Game-playing AI
- Robotics
- Autonomous vehicles
- Resource optimization
- Recommendation and decision systems

Key Point

Reinforcement Learning = Learning through actions, rewards, and penalties.

Difference Between Supervised, Unsupervised and Reinforcement Learning

Feature	Supervised Learning	Unsupervised Learning	Reinforcement Learning
Training Data	Labeled	Unlabeled	Interaction with environment
Correct Answer	Provided	Not provided	Reward/Penalty
Main Goal	Predict output	Find hidden patterns	Learn best actions
Learning Method	Learn from examples	Discover structure	Trial and error
Example	Spam detection	Customer grouping	Game-playing AI
Common Tasks	Classification, Regression	Clustering	Decision-making
Feedback	Correct/incorrect output	No direct feedback	Reward or penalty

Easy Real-Life Analogy

Supervised Learning → Teacher

A teacher gives you **questions along with correct answers**.

Question → Answer → Learn → Solve New Question

Unsupervised Learning → Finding Groups Yourself

You are given a box containing different objects but no labels. You examine them and **make groups based on similarities**.

Objects → Observe Similarities → Create Groups

Reinforcement Learning → Learning Through Experience

You learn to ride a bicycle by **trying, making mistakes, correcting yourself, and improving**.

Action → Result → Feedback → Improve → Try Again

Data, Data Visualization and Data Pre-processing

1. Data

Data refers to raw facts, observations, or information collected from different sources. Data can be used by computers and Machine Learning systems to **analyze information, identify patterns, and make predictions**.

Examples of Data

- Student marks: 75, 82, 68, 90
- Names: Rahul, Amit, Priya
- Images and videos
- Sensor readings
- Customer purchase records
- Social media posts
- Website activity

Types of Data

1. Structured Data:

Data organized in a fixed format such as rows and columns.

Example: Excel table or database.

2. Unstructured Data:

Data that does not follow a fixed tabular structure.

Example: Images, videos, audio, emails, and social media posts.

3. Semi-structured Data:

Data that has some organizational structure but does not follow a strict table format.

Example: JSON and XML files.

Data in Machine Learning

In Machine Learning, data is generally divided into:

Features (Input) → Model → Target (Output)

Example:

Study Hours + Attendance → ML Model → Pass/Fail

2. Data Visualization

Data Visualization is the process of representing data in a **visual form such as charts, graphs, and plots** so that information, patterns, trends, and relationships can be easily understood.

Instead of examining hundreds of numerical values, visualization allows us to understand the information quickly.

Common Types of Data Visualization

1. Bar Chart

Used to compare different categories.

Example: Number of students in different departments.

2. Line Chart

Used to show changes or trends over time.

Example: Monthly sales of a company.

3. Pie Chart

Used to show parts or percentages of a whole.

Example: Percentage of students using different programming languages.

4. Histogram

Used to understand the distribution of numerical data.

Example: Distribution of students' marks.

5. Scatter Plot

Used to show the relationship between two numerical variables.

Example: Study Hours vs Marks.

Importance of Data Visualization

- Makes complex data easier to understand
- Helps identify patterns and trends
- Helps detect outliers
- Supports better decision-making
- Makes comparison easier
- Communicates results effectively

Example

Suppose a college has the following marks:

45, 50, 55, 60, 65, 70, 75, 80

A graph can quickly show whether students' performance is generally **increasing, decreasing, or distributed around a particular range.**

3. Data Pre-processing

Data Preprocessing is the process of **cleaning, transforming, and preparing raw data before using it for analysis or Machine Learning.**

Real-world data is often incomplete, inconsistent, duplicated, or in an unsuitable format. Therefore, preprocessing is an important step before training an ML model.

Basic Process

Raw Data → Cleaning → Transformation → Feature Preparation → Final Dataset → ML Model

Major Steps of Data Preprocessing

1. Data Cleaning

Data cleaning involves finding and correcting problems in the dataset.

It may include:

- Handling missing values
- Removing duplicate records
- Correcting incorrect values
- Handling inconsistent data
- Removing unnecessary data

Example:

Name	Age
Amit	20
Rahul	—
Priya	21

The missing age value for Rahul needs to be handled before using the data.

2. Handling Missing Values

Missing values can be handled by:

- Removing the record
- Replacing it with **mean**
- Replacing it with **median**
- Replacing it with **mode**
- Using an appropriate prediction/imputation method

Example:

Ages:

20, 21, 22, -, 24

The missing value can be estimated using an appropriate statistical method.

3. Removing Duplicate Data

Sometimes the same record appears more than once.

Example:

Student	Marks
Amit	80
Rahul	75
Amit	80

The duplicate Amit record can be removed if it represents the same observation.

4. Data Transformation

Data may need to be converted into a suitable format.

Examples:

- Converting text into numerical values
- Changing date formats
- Converting units
- Encoding categorical values

Example:

Male → 0
Female → 1

This allows certain ML algorithms to work with categorical information.

5. Feature Scaling

Different features may have very different numerical ranges.

Example:

Age: 18–60

Salary: 20,000–2,00,000

Scaling brings numerical features to a comparable range.

Common methods include:

- **Normalization**
- **Standardization**

This can help certain ML algorithms perform better.

6. Handling Outliers

An **outlier** is a value that is significantly different from most other values.

Example:

20, 21, 22, 23, 24, 150

Here, **150** may be an outlier depending on what the data represents.

Outliers should be investigated because they may be caused by:

- Data entry errors
- Measurement errors
- Genuine unusual observations

7. Splitting the Dataset

After preprocessing, data is commonly divided into:

Training Data → Used to train the model

Testing Data → Used to evaluate the model

Sometimes a third set is used:

Validation Data → Used for model selection/tuning

A common example is:

Training: 80%

Testing: 20%

The exact split depends on the problem and dataset.

4. Relationship Between Data, Visualization and Preprocessing

These three concepts are closely connected in a Machine Learning workflow:

Data Collection



Data Preprocessing



Data Visualization / Exploration



Feature Preparation



Model Training



Prediction & Evaluation

Simple Example: Student Performance

Suppose a college collects:

Student Name + Attendance + Study Hours + Marks

Data: Raw student information.

Data Preprocessing:

Remove duplicates, handle missing marks, correct incorrect values, and prepare the features.

Data Visualization:

Create a graph of **Study Hours vs Marks** to understand whether more study generally relates to higher marks.

Machine Learning:

Use the prepared data to train a model that can predict the marks of a new student.

Quick Revision

Concept	Meaning	Main Purpose
Data	Raw facts and information	Provides information for analysis/ML
Data Visualization	Visual representation of data	Understand patterns and trends
Data Preprocessing	Cleaning and transforming raw data	Prepare quality data for analysis/ML

Characteristics of Artificial Intelligence

The major characteristics of AI describe the abilities that make a computer system behave intelligently.

1. Learning

AI systems can learn patterns from data and improve their performance.

Example: A spam filter learns which types of emails are usually spam.

2. Reasoning

AI can analyze available information and reach a logical conclusion.

Example: A medical AI can analyze symptoms and suggest possible diseases.

3. Problem Solving

AI can find solutions to problems by analyzing different possibilities.

Example: Google Maps finds an efficient route between two locations.

4. Decision Making

AI can select an appropriate action based on available information.

Example: A recommendation system decides which movie to recommend.

5. Perception

AI can understand information collected from the physical world through images, audio, sensors, etc.

Example: Face recognition systems identify a person from an image.

6. Natural Language Understanding

AI can process and understand human language.

Example: ChatGPT can understand questions written in natural language.

7. Adaptability

AI systems can adjust their behaviour when new data or situations are encountered.

8. Automation

AI can perform repetitive or complex tasks with reduced human intervention.

Advantages of Artificial Intelligence

Artificial Intelligence provides several benefits in different fields.

Major Advantages

1. Automation

AI can automate repetitive and time-consuming tasks.

2. Reduced Human Error

AI can reduce errors in tasks when properly designed and trained.

3. Fast Decision Making

AI can process large amounts of information quickly.

4. 24×7 Availability

AI-based systems can operate continuously without human fatigue.

5. Large-Scale Data Processing

AI can analyze huge amounts of data in a short time.

6. Increased Productivity

Automation and intelligent assistance can improve productivity.

7. Personalization

AI can provide personalized recommendations and services.

8. Dangerous Task Handling

Robots and AI systems can be used in dangerous environments such as mines, disaster areas, and space missions.

Disadvantages / Limitations of AI

Although AI has many advantages, it also has limitations and risks.

Major Disadvantages

1. High Cost

Developing, training, and maintaining advanced AI systems can be expensive.

2. Job Displacement

Automation may reduce the demand for humans in some repetitive jobs.

3. Lack of Human Emotions

AI does not naturally experience human emotions, empathy, or consciousness.

4. Dependence on Data

Poor-quality or biased data can produce poor or biased results.

5. Privacy Issues

AI systems may process large amounts of personal and sensitive information.

6. Security Risks

AI systems can be misused or attacked.

7. Bias

AI can reproduce or amplify biases present in its training data.

8. Lack of Common Sense

AI may perform well on specific tasks but can struggle with ordinary human reasoning and unfamiliar situations.

Applications of Artificial Intelligence

AI is used in almost every major industry.

Field	Applications
Healthcare	Disease detection, medical imaging, drug discovery
Education	AI tutors, personalized learning, automatic evaluation
Banking	Fraud detection, credit analysis, chatbots
Agriculture	Crop monitoring, disease detection, smart irrigation
Transportation	Route optimization, traffic prediction, autonomous vehicles
Manufacturing	Quality control, predictive maintenance, robotics
E-commerce	Product recommendations, customer support
Cybersecurity	Threat detection and anomaly detection
Entertainment	Movie, music and content recommendations
Business	Data analysis, forecasting, automation
Robotics	Industrial and service robots
Smart Cities	Traffic management, energy management and surveillance

Major AI Techniques and Technologies

AI is a broad field that includes several technologies and techniques.

1. Machine Learning (ML)

Machine Learning allows computers to learn patterns from data and make predictions or decisions.

2. Deep Learning (DL)

Deep Learning uses multi-layer neural networks to learn complex patterns, especially from large datasets.

3. Natural Language Processing (NLP)

NLP enables computers to understand, process, and generate human language.

Examples: Chatbots, translation, sentiment analysis.

4. Computer Vision

Computer Vision enables computers to understand and analyze images and videos.

Examples: Face recognition, object detection.

5. Robotics

Robotics deals with designing and controlling machines that can perform physical tasks.

6. Expert Systems

Expert systems use stored knowledge and rules to solve problems in a particular domain.

7. Speech Recognition

Speech recognition converts human speech into machine-readable text or commands.

Example: Voice assistants.

8. Knowledge Representation

Knowledge representation involves representing facts, concepts, and relationships in a form that an AI system can use for reasoning.

Artificial Intelligence Agents

An **AI Agent** is a system that **perceives its environment, processes information, and takes actions to achieve a specific goal.**

Basic Structure

Environment → Sensors → Agent → Actuators → Environment

Components

Sensors: Collect information from the environment.

Agent: Processes information and decides what action should be taken.

Actuators: Perform the selected action.

Example: Self-Driving Car

- **Sensors:** Cameras, GPS, radar and other sensors

- **Agent:** AI decision-making system
- **Actuators:** Steering, brakes and accelerator
- **Environment:** Road, vehicles, pedestrians and traffic signals

The AI agent continuously observes the environment and takes appropriate actions.

Key Point

An intelligent agent perceives its environment through sensors and acts upon that environment through actuators to achieve a goal.

AI Ethics

AI Ethics refers to the principles and practices used to ensure that AI systems are developed and used in a **fair, safe, transparent, and responsible manner**.

Major Principles of AI Ethics

1. Fairness

AI should not unfairly discriminate against individuals or groups.

2. Privacy

Personal data should be collected and used responsibly.

3. Transparency

Users should have an appropriate understanding of how an AI system reaches important decisions.

4. Accountability

People and organizations responsible for AI systems should be accountable for their use and outcomes.

5. Safety and Security

AI systems should be designed to reduce harmful outcomes and protect against misuse.

6. Human Control

Important decisions should have appropriate human oversight.

Example

Suppose an AI system is used for recruitment. If its training data contains historical bias, the AI may unfairly prefer certain candidates.

Therefore, **fairness and bias testing** are important in AI systems.

Challenges of Artificial Intelligence

AI development and deployment face several technical and social challenges.

Major Challenges

- **Data Quality:** AI needs reliable and relevant data.
- **Data Privacy:** Personal information must be protected.
- **Bias:** Biased training data can produce biased results.

- **Explain ability:** Some complex AI models are difficult to understand.
- **High Computational Cost:** Advanced AI models may require powerful hardware.
- **Security:** AI systems can be attacked or manipulated.
- **Reliability:** AI may produce incorrect results.
- **Ethical Issues:** AI can create questions about fairness and responsibility.
- **Lack of Common Sense:** AI may struggle with situations that are simple for humans.
- **Misuse:** AI can be used for harmful purposes such as misinformation or fraud.

AI, Automation and Robotics

These three concepts are related but different.

Artificial Intelligence

AI focuses on making machines capable of **intelligent behaviour, reasoning, learning, and decision-making**.

Example: AI system predicting traffic.

Automation

Automation means performing a task **automatically with minimal human intervention**.

Example: An automatic washing machine completing a washing cycle.

Automation does not necessarily require AI.

Robotics

Robotics deals with **physical machines that can sense, move, and perform actions in the real world**.

Example: A robotic arm assembling products in a factory.

Difference

AI	Automation	Robotics
Focuses on intelligence	Focuses on automatic task execution	Focuses on physical machines
Can exist without a physical robot	May work using simple rules	Usually involves physical hardware
Example: ChatGPT	Automatic billing system	Industrial robot

Relationship

AI + Robotics → Intelligent Robots

For example, a robot may use **Computer Vision + AI + Sensors** to identify objects and decide how to pick them up.

Generative AI

Generative AI is a type of AI that can **generate new content** based on patterns learned from training data.

It can generate:

- Text
- Images
- Audio
- Video
- Code

Examples

- ChatGPT → Text generation
- Image generation models → Images
- AI music tools → Music/audio
- Coding assistants → Computer code

Traditional AI vs Generative AI

Traditional AI:

Usually focuses on **classification, prediction, recommendation, or decision-making**.

Generative AI:

Focuses on **creating new content**.

Example:

Traditional AI: "Is this email spam?" → **Spam**

Generative AI: "Write an email to my teacher." → **Generates an email**

Large Language Models (LLMs)

A **Large Language Model (LLM)** is an AI model trained on large amounts of text data to understand and generate human-like language.

LLMs can perform tasks such as:

- Question answering
- Text generation
- Summarization
- Translation
- Code generation
- Information extraction
- Conversational interaction

Example

User: Explain Machine Learning in simple words.

LLM: Generates an explanation based on patterns learned during training.

Relationship

AI → Machine Learning → Deep Learning → Neural Networks → Large Language Models
→ Generative AI Applications

This is a simplified conceptual relationship; not every AI system follows this exact chain.

Where Does AI Stand Today?

Artificial Intelligence has evolved from simple rule-based systems to advanced systems that can **understand text, generate content, write software, analyze images and videos, use tools, operate computers, and control robots.**

Today, AI is moving from simply **answering questions** to **performing complete tasks.**

A simple evolution can be understood as:

AI → Machine Learning → Deep Learning → Generative AI → AI Agents → Physical AI/Robotics

1. AI Can Generate Text

Modern Generative AI systems can understand natural-language instructions and generate human-like text.

AI can generate:

- Articles
- Notes
- Emails
- Reports
- Stories
- Summaries
- Questions and answers
- Computer documentation
- Business content

Example:

A student can give a prompt:

"Explain Machine Learning for B.Tech students in simple English with examples."

The AI can generate structured study material within seconds.

Technology: Generative AI + Large Language Models (LLMs)

2. AI Can Generate Images

Modern AI systems can create images from **text descriptions**, a process called **text-to-image generation.**

Example:

Prompt:

"Create an image of a futuristic smart city with autonomous cars and humanoid robots."

The AI can generate a new image based on the description.

Applications

- Advertising
- Graphic design
- Marketing
- Education
- Game development
- Product visualization
- Social media content

Technology: Generative AI + Diffusion/Generative Models

3. AI Can Generate and Edit Video

AI has moved beyond static images and can now generate or modify video content from text, images, or other video inputs.

Examples of tasks

- Text-to-video generation
- Image-to-video animation
- Video editing
- Background replacement
- Character animation
- AI-generated advertisements

This means a creator can describe a scene in natural language and use AI to produce a video concept without manually creating every frame.

4. AI Can Understand Images and Videos

AI is no longer limited to text.

Modern **multimodal AI** can work with:

Text + Images + Audio + Video

For example, an AI system can receive a photograph and answer:

"What objects are visible in this image?"

It can also analyze documents, charts, screenshots, diagrams, and photographs.

Technology: Computer Vision + Multimodal AI

5. AI Can Understand and Generate Voice

Modern AI can perform both:

Speech → Text

and

Text → Speech

This enables AI systems to communicate using natural human language.

Applications

- Voice assistants
- AI customer support
- AI tutors
- Voice translation
- Accessibility tools
- Voice-based agents

For example:

User speaks → AI understands → AI reasons → AI generates answer → AI speaks

6. AI Can Write Computer Programs

One of the biggest recent developments is **AI-assisted software development**.

AI can now help developers:

- Write code
- Explain code
- Find bugs
- Fix errors
- Generate tests
- Refactor code
- Create APIs
- Create websites
- Create database structures
- Write documentation

For example:

"Create a React website for a college management system with login, dashboard and student records."

An AI coding system can generate a substantial part of the application.

7. Vibe Coding

Vibe Coding is a modern approach to software development where a person describes what they want in **natural language**, and AI generates much of the required code.

Instead of manually writing every line:

Idea → Natural Language Prompt → AI Generates Code → Test → Improve

Example

A user says:

"Build a modern portfolio website with a dark theme, animations, projects section and contact form."

An AI coding tool can create the initial website, modify it based on feedback, and help debug it.

Important Point

Vibe coding does **not mean that programming knowledge is no longer useful**. Developers still need to understand:

- Requirements
- Architecture
- Security
- Testing
- Debugging
- Databases
- APIs
- Code quality

AI can generate code, but **human verification remains important**.

8. AI Coding Agents

A traditional AI coding assistant generally helps you **write code**.

An **AI Coding Agent** can go further.

It can:

Understand Task → Plan → Write Code → Modify Files → Run Tests → Debug → Report Results

Examples of AI coding agents include **Devin, Replit Agent, Claude Code, Codex and other agentic coding systems**.

Example

Instead of saying:

"Write a login function."

You can give an agent:

"Add Google login to my React application. Update the backend, database configuration and frontend, run the tests, fix any errors, and prepare the changes for review."

The agent can perform multiple steps rather than simply returning a code snippet.

9. Real-World Example: Devin

Devin is an example of an AI software-engineering agent developed by Cognition.

The important idea behind Devin is that it can work on software tasks more autonomously than a normal code-completion tool.

A coding task can follow:

Task → Planning → Coding → Testing → Debugging → Pull Request

Modern AI coding agents are increasingly being tested on real software-development workflows rather than only simple code-generation demonstrations.

Example Task

"Fix this bug, add a regression test, run the test suite, and create a pull request."

This represents a shift from:

AI as a Coding Assistant

to

AI as a Software Engineering Agent

10. Real-World Example: Replit Agent

Replit has developed AI agents that allow users to describe an application in natural language and have AI assist with building and deploying it.

Replit reported in 2026 that its internal agents were being used for tasks including **investigating production incidents, reviewing pull requests, analyzing business data, triaging support tickets, and researching sales accounts.**

This demonstrates that AI agents are moving beyond simple code generation toward **complete workplace tasks.**

11. AI Can Use Computers

Another major development is **Computer-Using AI Agents.**

Instead of only generating text, an AI agent can potentially interact with a computer through:

- Mouse
- Keyboard
- Browser
- Applications
- Web pages
- Files

Example

A user could ask an agent:

"Find the required information on this website, put it into a spreadsheet, and prepare a summary."

The agent can potentially navigate websites and perform multiple steps.

OpenAI and other AI developers are actively developing computer-use capabilities, although reliability, security, privacy, and generalization across websites remain important challenges.

12. AI Agents

Traditional AI:

User asks → AI answers

Agentic AI:

User gives goal → AI plans → AI uses tools → AI performs actions → AI checks results

This is known as **Agentic AI** or **AI Agents**.

Example

Suppose you say:

"Research the best laptops under ₹60,000, compare their specifications, and prepare a table."

A simple AI might give you information.

An AI agent could potentially:

Search → Collect Information → Compare → Analyze → Prepare Result

This is a major direction in current AI development.

13. AI in Robotics – Physical AI

AI is also moving from the **digital world into the physical world**.

This is sometimes called **Physical AI** or **Embodied AI**.

A robot can combine:

AI + Computer Vision + Sensors + Robotics + Planning

to understand its surroundings and perform physical actions.

Google DeepMind's **Gemini Robotics** work is an example of this direction: its vision-language-action models are designed to translate visual information and instructions into actions for robots.

14. AI-Powered Football/Soccer Robots

A very interesting real-world example is **Boston Dynamics' Atlas humanoid robot**.

In 2026, Hyundai and Boston Dynamics demonstrated Atlas performing football-related tasks such as:

- Dribbling
- Passing
- Shooting
- Complex movement

The demonstration shows how AI-powered humanoid robots are being developed to perform increasingly complex physical tasks.

Concept

See the ball → Understand position → Plan movement → Move body → Kick ball

This is much more complex than simply generating text because the AI has to interact with the **physical world**.

15. AI Robots Are Becoming More General

Recent robotics research is moving toward robots that can learn **new physical tasks** rather than being programmed for only one fixed action.

For example, research demonstrations in 2026 showed robots learning tasks from short instructions or videos and adapting when the situation changed. However, these systems are still imperfect and are far from human-level general-purpose robotics.

This is an important distinction:

AI is becoming more capable, but today's AI is still not equivalent to human general intelligence.

16. AI in Content Creation and Marketing

AI is already being used in real business workflows for:

- Product images
- Advertisements
- Marketing copy
- Social media content
- Video creation
- Campaign optimization
- Customer communication

For example, Reuters reported in 2026 that companies including **Kimberly-Clark, Catalyst Brands and Target India** were using AI in advertising and marketing workflows. Kimberly-Clark reported reducing content creation time from **24 days to about two hours** using an AI platform.

This is a good example of AI moving from **experimentation to business operations**.

17. AI in Social Media

AI is deeply integrated into social media platforms.

It can be used for:

- Content recommendation
- Image/video understanding
- Spam detection
- Content moderation
- Advertising
- Personalization
- Content generation
- Trend analysis

For example, AI systems can analyze millions of pieces of content and determine what content may be relevant to a particular user.

Basic Process

User Activity → AI Analysis → User Interest Prediction → Personalized Feed

18. AI Can Perform Multiple Tasks Together

Modern multimodal AI systems can combine different capabilities.

For example, a single system may be able to:

Read a PDF → Understand a diagram → Analyze data → Write a report → Generate code

This is different from older AI systems that were generally designed for **one specific task**.